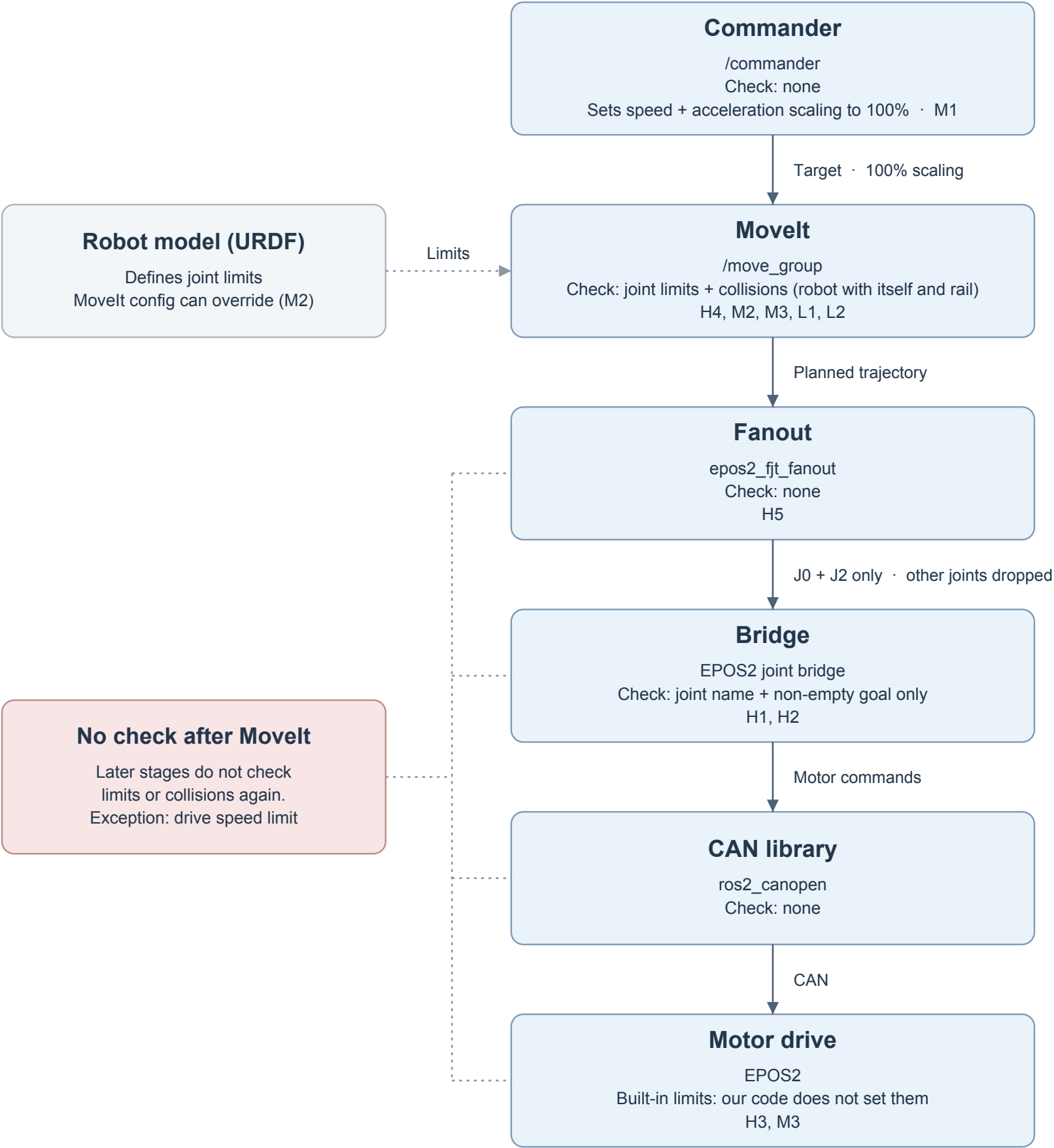


# Joint limit and collision checks along the move chain



| Joint limits (reference) |                |       |         |            |
|--------------------------|----------------|-------|---------|------------|
| Joint                    | Motion         | Min   | Max     | Max speed  |
| J0                       | Rail           | 0 m   | 1.565 m | 1.0 m/s *  |
| J1                       | Shoulder turn  | -180° | +180°   | 0.40 rad/s |
| J2                       | Shoulder pitch | -90°  | +90°    | 1.96 rad/s |
| J3                       | Elbow          | -258° | +86°    | 3.65 rad/s |
| J4                       | Wrist 1        | -180° | +180°   | 2.00 rad/s |
| J5                       | Wrist 2        | -286° | +286°   | 3.50 rad/s |
| J6                       | Wrist turn     | 0°    | +120°   | 1.00 rad/s |
| Fins                     | Gripper        | -14°  | +10°    | 1.00 rad/s |

\* J0 max speed: Movelt value. Robot model value: 0.5 m/s (M2).

# Joint limit and collision check issues by severity

| Issue                                                               | Affects        | Problem                                                                                                                                                                                                                                                                                                        | Consequence                                                                                                                     | Proposed fix (not done)                                                                                               |
|---------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| HIGH SEVERITY 5 issues                                              |                |                                                                                                                                                                                                                                                                                                                |                                                                                                                                 |                                                                                                                       |
| <b>H1 J2–J6 are not homed</b><br>High                               | Bridge, drives | Joint configs set homing_enabled: false and allow_unhomed_motion: true. No code reads these flags. The bridge has no homing code. ros2_canopen can home a drive, but no code calls it. The README calls homing optional.                                                                                       | Zero can be at a different position after each power-on. All joint limits can then point at wrong angles.                       | Add homing (home switch or absolute encoder). Block bridge motion until homing is complete. Needs EE/ME for hardware. |
| <b>H2 Bridge does not check position or speed</b><br>High           | Bridge         | The goal check accepts a matching joint name and a non-empty goal. It does not check the target position or the speed.                                                                                                                                                                                         | Any ROS program (test script, typo, bug) can send a bad target directly to the motor.                                           | Add min/max position checks and a max speed check in the bridge.                                                      |
| <b>H3 Our code does not set drive limits</b><br>High                | Drives (EPOS2) | Code does not set position (0x607D), speed (0x607F), acceleration (0x60C5), or following-error (0x6065) limits. Profile speed/accel (0x6081/0x6083) only give a warning. Saved file (one drive, March): position limit off, acceleration limit off, speed limit 1960 rpm (active: the drive faults and stops). | The drive is the last safety net. With the position limit off, the drive goes to any commanded position.                        | Read the settings from each drive. Then set real limits. Needs EE.                                                    |
| <b>H4 MoveIt does not know the environment</b><br>High              | MoveIt         | No code adds obstacles. MoveIt knows only the robot and the rail. Missing: floor, plants, wall, bin, camera mount, electronics box.                                                                                                                                                                            | MoveIt can plan a path through the floor or the plants.                                                                         | Add the floor, plants/wall, bin, and mounts at startup. Needs measurements from ME.                                   |
| <b>H5 MoveIt checks one path, the robot follows another</b><br>High | Fanout         | The fanout in this repo moves only J0 and J2. For a 7-joint goal, it drops the other 5 joints and logs a warning. The README says that arm_trajectory_fanout (NucBox) is the daily fanout. Its behavior is not known.                                                                                          | The collision check applies to a path that the robot does not follow.                                                           | Set allow_extra_goal_joints to false, or plan only with the connected joints.                                         |
| MEDIUM SEVERITY 3 issues                                            |                |                                                                                                                                                                                                                                                                                                                |                                                                                                                                 |                                                                                                                       |
| <b>M1 Commander runs at 100% speed</b><br>Medium                    | Commander      | The MoveIt config default is 10%. The commander overrides it to 100%.                                                                                                                                                                                                                                          | Every move runs at full speed: J3 up to 3.65 rad/s (~209 deg/s), rail up to 1.0 m/s.                                            | Use 10–20% during tests.                                                                                              |
| <b>M2 Rail (J0) speed limit is double the model value</b><br>Medium | MoveIt config  | Robot model: 0.5 m/s. MoveIt config: 1.0 m/s (changed May 9. The backup from that day has 0.5). MoveIt uses 1.0 m/s.                                                                                                                                                                                           | The rail carries the whole arm. The stopping distance at 1 m/s is not known.                                                    | Ask ME for the max speed. Measure the stopping distance.                                                              |
| <b>M3 MoveIt speeds exceed the motor speed limit</b><br>Medium      | MoveIt, drives | The saved drive file caps the motor at 1960 rpm. After the gearbox, MoveIt limits need J2 3144 rpm, J3 3485 rpm, J5 3342 rpm (too fast). J4 (1910 rpm) and J6 (955 rpm) are OK.                                                                                                                                | At 100% speed, a long fast move can exceed the cap. The drive then faults and the arm stops mid-move.                           | When the team knows the real motor max speed, lower those joint speeds to match.                                      |
| LOW SEVERITY 2 issues                                               |                |                                                                                                                                                                                                                                                                                                                |                                                                                                                                 |                                                                                                                       |
| <b>L1 gripper_closed pose is out of range</b><br>Low                | MoveIt SRDF    | The pose sets fin_joint1 to 1.0. The max is 0.175. No code uses the pose yet (no gripper controller).                                                                                                                                                                                                          | No effect today.                                                                                                                | Use a value from -0.25 to 0.175.                                                                                      |
| <b>L2 Some link pairs skip collision checks</b><br>Low              | MoveIt SRDF    | Skipped pairs: rail/link2, rail/link3, base/link2, base/link3, rail/link1. A check with the real collision shapes over the full joint range found no contact. Closest: link3 to rail, 16 mm, at J2 = +90°.                                                                                                     | Acceptable for now. The 16 mm margin holds only if the J2 limit is real (needs homing, H1) and the real rail matches the model. | Check again after the team adds homing.                                                                               |